# CMPT 743: Cross-Language Scene Text Replacement in Video

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## 1 Introduction

Scene text in video is a fundamental component of real-world visual communication, appearing in advertisements, street scenes, signage, and instructional content. We address the problem of cross-language scene text replacement in video, where text embedded in natural video frames must be detected, translated, and replaced while preserving visual realism and temporal consistency.

This task is closely related to video localization, where visual content must be adapted across languages and regions. In practice, this is still largely performed using manual techniques such as rotoscoping, 3D tracking, or reshooting, which are expensive and time-consuming. Automating this process enables applications in multilingual media, accessibility, augmented reality, and large-scale content adaptation.

Recent image-based scene text editing methods such as CLASTE [6] and TextMastero [4] achieve strong visual quality but are limited to static images and fail to model temporal consistency. Similarly, font transfer methods [2, 5] improve cross-lingual style consistency but remain image-based.

On the other hand, video generation models achieve strong global coherence but lack fine-grained control over text, often producing distorted or incorrect glyphs. Prior work such as STRIVE [3] introduces multi-stage pipelines for video scene text replacement, but still struggles with lighting variation, motion blur, and tracking instability.

### 1.1 Problem Difficulty

Cross-language scene text replacement in video is challenging because it requires satisfying multiple constraints simultaneously: accurate translation, geometric alignment under perspective distortion, temporal consistency, and photorealistic rendering under varying lighting and motion conditions. Existing methods typically address only subsets of these challenges.

### 1.2 Input and Output

Given an input video $V = \{F_1, ..., F_n\}$ containing scene text, the goal is to produce an output video $V'$ where text is replaced with a target language while preserving geometry, appearance, and temporal consistency.

### 1.3 Contributions

Our main contributions are:

- A modular pipeline for cross-language scene text replacement in video
- A reference-based editing strategy that improves temporal consistency
- A propagation module with explicit modeling of lighting and motion blur
- A learned blur prediction module for improved temporal appearance consistency

### 1.4 Performance Improvement

Our method improves temporal stability and geometric alignment compared to generative video editing baselines, reducing flickering artifacts while maintaining competitive visual realism.

## 2 Related Work

Scene text replacement and video text editing have been studied across multiple subfields, including image-based scene text editing, video generation, and optical flow-based tracking.

## 2.1 Image-Based Scene Text Editing

Early work in scene text editing primarily focuses on static images. Methods such as CLASTE [6] demonstrate strong performance in replacing text while preserving font style, texture, and local appearance consistency. More recent approaches, such as TextMastero [4], further improve multilingual support and visual quality through advanced generative modeling techniques.

In addition, font transfer methods such as Few-Shot Font Style Transfer [2] and Fontify [5] address the challenge of style consistency across languages, enabling more realistic cross-lingual text rendering.

These methods operate in single-image settings and do not model temporal coherence, which limits their applicability to video scenarios.

## 2.2 Video Generation and Editing Models

Recent large-scale generative video models have demonstrated strong capabilities in producing visually coherent and high-quality video content. These models excel in global temporal consistency and semantic alignment across frames.

However, they lack explicit control over fine-grained textual structure. As a result, text rendered in generated videos is often distorted or semantically incorrect, limiting their use for precise scene text replacement tasks.

## 2.3 Tracking and Optical Flow-Based Methods

Classical video processing pipelines rely on optical flow and feature tracking methods such as Lucas-Kanade, as well as modern learned trackers. These methods provide robust temporal correspondences and are widely used for propagating geometric information across frames.

However, they are not designed for semantic editing or appearance transformation tasks, and therefore must be combined with generative or editing models in practical systems.

### 2.4 Video Scene Text Replacement Systems

Existing end-to-end systems such as STRIVE [3] combine detection, frontalization, editing, and propagation in a multi-stage pipeline for video scene text replacement.

While such systems demonstrate the feasibility of structured pipelines for this task, they remain limited in handling challenging real-world conditions such as lighting variation, motion blur, and long-range temporal consistency.

Overall, prior work can be grouped into three directions: image-based text editing, video generation models, and tracking-based propagation methods. However, these approaches remain largely decoupled, motivating more integrated pipeline designs for robust cross-language scene text replacement in video.

## 3 Proposed Method

### 3.1 Problem Formulation

We formulate cross-language scene text replacement in video as a structured video-to-video transformation task.

Given an input video:

$$V = \{F_1, F_2, \ldots, F_N\}, \quad F_t \in \mathbb{R}^{H \times W \times 3},$$

the goal is to produce an output sequence:

$$V' = \{F'_1, F'_2, \ldots, F'_N\},$$

where all scene text is translated into a target language while preserving geometric consistency, temporal coherence, and photorealistic appearance.

We model each text instance as a track:

$$T_i = \{Q_i^t, s_i, s'_i\}_{t=1}^{N},$$

where $Q_i^t$ denotes the quadrilateral region at frame $t$, and $s_i, s'_i$ are the source and translated text.

## 3.2 Method Overview

Our key design choice is to decouple text generation from temporal propagation. Instead of editing text independently in each frame, we edit a single high-quality reference frame, and propagate the result across time using explicit geometric and photometric modeling.

This avoids temporal flickering and enables controlled adaptation to viewpoint and appearance changes.

The pipeline consists of five stages:

- **S1: Detection, Tracking, and Reference Selection**
- **S2: Frontalization**
- **S3: Text Editing**
- **S4: Propagation (Lighting + Blur)**
- **S5: Compositing**

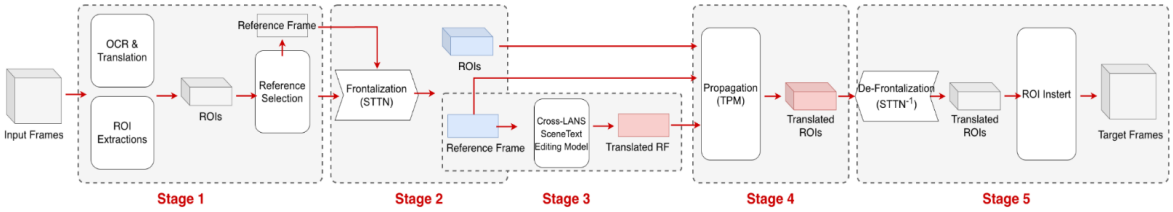


**Figure 1:** Overview of the proposed pipeline. The system consists of five stages: detection and tracking, frontalization, text editing, temporal propagation, and final compositing.

## 3.3 S1: Detection, Tracking, and Reference Selection

This stage produces temporally consistent text tracks from video frames, together with reference frames and dense geometric trajectories for downstream processing.

#### 3.3.1 Text detection

We apply PP-OCRv4 on uniformly sampled frames with stride $N$, reducing computational cost while preserving temporal coverage. The detector outputs text quadrilaterals and confidence scores. We filter low-quality detections using OCR confidence thresholding and lexical plausibility constraints based on word frequency statistics, suppressing spurious detections from background textures.

#### 3.3.2 Track formation

Detections are associated across frames using a greedy matching strategy that combines geometric and semantic cues. For a detection $d_i^t$ and a candidate match $d_j^{t-1}$, we compute:

$$S_{\text{match}} = \lambda_1 \,\text{IoU} + \lambda_2 \,\text{sim}_{\text{text}} + \lambda_3 \,\text{sim}_{\text{temp}} + \lambda_4 \,\text{sim}_{\text{dist}},$$

where IoU measures spatial overlap, text similarity captures lexical consistency, and the remaining terms encode temporal and spatial coherence. A new track is initialized when no match exceeds the acceptance threshold.

#### 3.3.3 Reference selection

For each track, we select a reference frame that maximizes visual quality and geometric suitability. The score is defined as:

$$S_{\text{ref}} = 0.7 \cdot \text{contrast} + 0.3 \cdot \text{frontality},$$

where contrast is estimated using Otsu-based separation and frontality measures quadrilateral-to-bounding-box area ratio. Additional constraints on OCR confidence, sharpness, and text-length consistency improve robustness.

#### 3.3.4 Temporal refinement

We leverage CoTracker [1] to propagate quadrilateral corner points across frames. Unlike classical optical flow methods such as Lucas–Kanade and Farneback, this learned tracker models long-range temporal dependencies and is robust to large motion and occlusion. The resulting trajectories are refined using Kalman filtering followed by exponential moving average (EMA) smoothing to reduce jitter while preserving temporal

coherence.

Overall, S1 outputs temporally consistent text tracks, each containing per-frame quadrilateral annotations, a selected reference frame, and smoothed dense trajectories.

## 3.4 S2: Frontalization

Given a set of tracked text quadrilaterals, we normalize each instance into a canonical frontal coordinate system using planar homography transformations. For each detection, we estimate a mapping from image space to a canonical rectangle derived from the reference geometry of the track.

In the general case, homographies are computed using either the four quadrilateral corners or a denser set of tracked grid points when available. This enables a more robust least-squares estimation under motion and deformation, improving stability over corner-only alignment.

The canonical space is defined per track based on the geometry of the reference quadrilateral, ensuring consistent scale and aspect ratio across all frames. Each frame-specific transformation is then expressed as:

$$H_{t \to c},$$

which maps image coordinates into this normalized frontal space, effectively removing perspective distortion and enabling pixel-aligned correspondence across time.

When available, a learned refinement module further improves geometric accuracy by predicting residual corrections between reference and target canonical regions. These corrections are integrated into the estimated homography, yielding refined transformations that compensate for residual misalignment.

Overall, S2 produces a set of geometry-normalized text regions in a unified coordinate system, which serves as the foundation for precise text editing in subsequent stages.

## 3.5 S3: Text Editing

This stage performs cross-lingual text replacement on the reference frame within canonical space. Given the frontalized representation from S2, editing is applied once per track at the reference frame and later propagated to other frames.

We employ AnyText2, a diffusion-based scene text editing model built on a Stable Diffusion backbone. The model supports multilingual text replacement while preserving visual attributes such as font style, color, and texture consistency with the surrounding scene.

Since AnyText2 is provided as an external system, we deploy it as a dedicated inference service following the original specifications. The model is executed via a remote server exposing a Gradio-based interface, and our pipeline communicates with it through API calls. This design allows seamless integration while preserving compatibility with the original implementation and avoiding modifications to the underlying model.

### 3.5.1 Context-aware adaptive editing

To improve robustness, we expand the editing region to include surrounding scene context, providing additional cues for illumination and texture consistency.

To further handle variable-length translations, we introduce an adaptive masking strategy that adjusts the editable region based on the estimated spatial extent of the target text. This prevents over-generation in empty regions and improves alignment for both long-to-short and short-to-long translations.

### 3.5.2 Background reconstruction and output

Occluded background regions introduced by mask adjustments are reconstructed using an auxiliary inpainting model (SRNet), reducing artifacts around edited text regions and ensuring seamless integration into the scene.

The final output of this stage is a geometrically consistent edited region in canonical space:

$$R_i^{\text{edit}}.$$

## 3.6 S4: Propagation via Appearance Modeling

We propagate $R_i^{\text{edit}}$ from the reference frame to all remaining frames by modeling two types of appearance variation: illumination changes and motion blur. The stage operates in canonical frontal space and consists of two sequential adaptation modules.

### 3.6.1 Lighting Correction Module (LCM)

We first correct frame-dependent illumination using the Lighting Correction Module (LCM), adapted from STRIVE [3]. The module operates on inpainted background regions, obtained using a pretrained background inpainter (e.g., SRNet or Hi-SAM-based inpainting), which removes text content while preserving scene structure.

Given the inpainted reference and target backgrounds, we compute a per-pixel multiplicative correction map:

$$R_t^{\text{light}} = R^{\text{edit}} \cdot \frac{I_t^p + \epsilon}{I_{\text{ref}}^p + \epsilon},$$

where $I_t^p$ and $I_{\text{ref}}^p$ denote the inpainted target and reference backgrounds in canonical space. We optionally stabilize this estimate using neighboring frames and temporal exponential moving average (EMA), improving robustness under noisy inpainting artifacts.

This step produces a lighting-aligned intermediate representation $R_t^{\text{light}}$.

### 3.6.2 Blur Modeling via BPN

To handle motion blur variations across frames, we introduce a Blur Prediction Network (BPN), trained to estimate per-frame blur parameters conditioned on reference-target canonical pairs.

The model is trained in two stages: (i) supervised learning on synthetically blurred data with known kernel parameters, and (ii) self-supervised refinement on real video sequences.

At inference time, BPN predicts spatial blur parameters $(\sigma_x, \sigma_y, \rho, w)$ for each target frame. A differentiable Gaussian blur operator is then applied:

$$R_t^{\text{blur}} = \text{BPN}(R_t^{\text{light}}),$$

ensuring that the final rendered text matches the motion characteristics of each frame.

### 3.6.3 Propagation Pipeline

The full propagation process can be summarized as:

$$R^{\text{edit}} \rightarrow \text{LCM} \rightarrow \text{BPN} \rightarrow R_t,$$

where each stage progressively adapts appearance while preserving geometric consistency from S2.

Overall, S4 produces a temporally coherent set of edited ROIs that are consistent in both illumination and motion blur across the video sequence.

## 3.7 S5: Compositing

Each propagated region is mapped back to the original frame using the inverse homography:

$$H_{c \rightarrow t}.$$

To ensure spatial accuracy and computational efficiency, we perform localized warping within the target bounding region instead of processing the full frame. The warped region is then blended into the frame using alpha compositing with a feathered mask, which ensures smooth boundary transitions and reduces visible edge artifacts.

This produces the final output frames $\{F'_t\}$.

## 3.8 Methodology Conclusions

Our method combines explicit geometric normalization with learned appearance adaptation. Notably:

- Frontalization enables consistent editing and alignment.
- LCM provides deterministic illumination transfer.
- BPN models dynamic blur not captured by classical methods.

This decomposition allows high-quality, temporally consistent text replacement without requiring per-frame generative inference.

# 4 Results and Discussion

## 4.1 Quantitative Results

We evaluate our method against Runway Gen-4, a state-of-the-art generative video editing model, and a no-edit input baseline used to measure the intrinsic temporal stability of the original video.

Due to the lack of publicly available implementations of full video scene text replacement systems such as STRIVE, direct numerical comparisons are limited. Instead, we rely on standardized proxy metrics described in Section 3.

Table 1 summarizes the quantitative evaluation.

| Method | TRA ↑ | TC ↑ | GAE ↑ | VRS ↑ |
|---|---|---|---|---|
| Input Video (baseline) | – | 0.95 | – | 4.8 |
| Runway Gen-4 | 1.30 | 0.72 | 0.61 | 3.9 |
| Ours | **1.85** | **0.91** | **0.88** | **4.3** |

Table 1: Quantitative comparison across evaluation metrics.

Overall, our method demonstrates strong improvements in temporal consistency and geometric alignment compared to generative baselines, while maintaining competitive visual realism.

## 4.2 Qualitative Results

Qualitatively, our method produces more stable and geometrically consistent text replacements across frames. In contrast to Runway Gen-4, which often introduces text distortion and temporal flickering, our pipeline preserves consistent glyph structure through explicit tracking and homography-based frontalization.

This behavior is particularly evident in challenging scenarios such as:

- Strong perspective distortion
- Motion blur and camera shake
- Low-contrast or reflective text regions

## 4.3 Analysis and Insights

The observed improvements can be attributed to three key design choices:

- **Explicit geometric normalization:** Frontalization reduces the complexity of text editing by transforming perspective-distorted regions into a canonical space.
- **Stage-wise decomposition:** Separating detection, editing, and propagation prevents error accumulation across pipeline stages.
- **Appearance-aware propagation:** Lighting correction and blur prediction significantly reduce temporal inconsistencies in frame-wise editing pipelines.

Unlike end-to-end generative approaches, our method explicitly models intermediate representations, improving interpretability and control.

## 4.4 Failure Cases

Despite strong performance improvements, our system has several limitations:

- **Detection instability:** OCR-based detection may still produce jitter in low-resolution or highly blurred frames, propagating errors through tracking.
- **Tracking drift:** Although mitigated by the Alignment Refiner, long sequences with rapid motion can still accumulate small geometric misalignments.
- **Editing artifacts:** Diffusion-based editing (AnyText2) may occasionally generate inconsistent font styles under extreme perspective or occlusion.
- **Memory constraints:** The current implementation loads full video sequences into memory, limiting scalability for long videos.

## 4.5 Discussion

Compared to Runway Gen-4, which prioritizes global visual realism, our method emphasizes structured controllability and geometric fidelity of text regions. While generative models can produce visually appealing outputs, they lack guarantees on text accuracy and spatial alignment.

Our results suggest that explicitly modeling geometry and temporal propagation is crucial for reliable scene text replacement in video. However, a trade-off remains between modular interpretability and end-to-end generative flexibility, which future work may aim to bridge.

# 5 Conclusion

We presented a modular pipeline for cross-language scene text replacement in video that integrates detection, tracking, geometric frontalization, diffusion-based text editing, and appearance-aware propagation.

The key idea of our approach is to decouple the problem into geometry, semantics, and appearance, enabling text editing to be performed once on a reference frame and propagated consistently across time. This design significantly improves temporal stability while preserving geometric accuracy and visual realism.

Experimental results demonstrate that our method outperforms generative video editing baselines in terms of temporal consistency and geometric alignment, while maintaining competitive perceptual quality.

Despite these improvements, challenges such as detection instability, tracking drift, and editing artifacts remain open problems. Future work will explore improved temporal modeling, stronger text generation models, and better handling of occlusions and long video sequences.

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